

# NATURAL STRICT FILLERS DO NOT EXIST: A NO-GO THEOREM FOR DE MORGAN CUBICAL SETS AND THE MEDIAN CORRECTION STRUCTURE

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ABSTRACT. Over the free De Morgan algebras that govern the strict layer of CCHM cubical type theory, a boundary does not determine its filler: the fibers of the boundary map are Boolean lattices  $2^D$ , and a companion paper showed that the canonical minimal filler fails to commute with substitution in a controlled way. This paper closes the question of whether the failure can be repaired by a cleverer choice: it cannot. *No section of the boundary-fiber correspondence — no rule assigning to every boundary one of its strict fillers — is natural with respect to substitution*; the proof takes three lines from one uniquely-fillable boundary whose transports under a diagonal and under a connective collapse land at the top and the bottom of a single two-element fiber, and an exhaustive machine search certifies that the obstruction is ubiquitous rather than exceptional. What replaces the impossible choice is *structure*: the median interpolation of the companion papers is computed in fiber coordinates to be a graded Boolean median, its compositional coherence is witnessed by explicit *iterated* median cylinders whose faces match on the nose, and every compatible sphere of corrections fills — the system of corrections is strictly contractible. As a consequence, any filling operation for De Morgan cubical sets that is natural in the cube category must take values outside the strict layer: the weak (Kan) layer is categorically necessary, not merely convenient.

## 1. INTRODUCTION

This paper resolves the last open question of a structure theory, developed in [7, 8, 9], of the *strict layer* of De Morgan cubical sets — the presheaves on the cube category  $\square_{\text{DM}}$  of the CCHM interval theory [1], restricted to the reparametrizations that a proof-assistant kernel applies judgmentally. The theory so far: boundary fibers in the free De Morgan algebra  $\text{DM}_n$  are Boolean intervals  $2^D$  over antichains of diagonal points, Sperner-extremal in size [7]; every fiber has a canonical least element, reachable by the one-step *defect formula*; and the least element commutes with a substitution  $f$  exactly when the transported boundary avoids the defect antichain —

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2020 *Mathematics Subject Classification*. 06D30; 18N40, 55U35, 03B38.

*Key words and phrases*. cubical sets, De Morgan algebra, filling, algebraic weak factorization system, median, naturality, cubical type theory.

the *coherence dichotomy* [8]. The minimal filler is thus not natural. The question this paper answers:

*Is any choice of strict fillers natural? Or is the failure of  
naturality intrinsic to the strict layer?*

The answer is the strongest possible no-go.

**Theorem** (Theorem 3.1). There is no section of the boundary-fiber correspondence on the free De Morgan algebras — no assignment  $s$  of a filler  $s(B) \in \text{fib}(B)$  to every boundary  $B$  — such that  $s(\partial(f^*g)) = f^*(s(B))$  for all substitutions  $f$  and all  $g = s(B)$ . Already at two generators, and already for endomorphisms of [2], the constraints are unsatisfiable.

The proof is three lines (§3): the element  $m = \neg x \wedge y \wedge \neg y$  is the *unique* filler of its boundary, so any natural  $s$  is forced on it and on all its transports; but the diagonal  $x := y$  and the connective collapse  $x := x \wedge \neg x$  send  $m$  to the top, respectively the bottom, of one and the same two-element fiber — two forced values for one boundary. The two substitutions realize the two faces of the defect formula: one *adds* the invisible diagonal point, the other does not. A machine census shows this is no isolated accident: unique fillers force values on 42 of the 54 non-singleton fibers of  $\text{DM}_2$ , with 142 pairwise conflicts, and an exhaustive search over all  $2^{38} \cdot 4^{16}$  sections, constrained by 121 endo-substitutions, confirms unsatisfiability without using higher cubes.

The no-go sharpens the slogan of [8] — *without Eilenberg–Zilber normal forms, thinness cannot be a property* — into a theorem about *choices*: not only is there no property singling out a filler, there is no natural *choice function* either. What survives is the correction *structure*, and the second half of the paper (§4) shows it is as good as such a structure can be. In fiber coordinates the median interpolation  $\chi = (\varphi \wedge \neg k) \vee (k \wedge \varphi') \vee (\varphi \wedge \varphi')$  of [7] is precisely a graded Boolean median on  $2^D$  (Proposition 4.1); the composition of corrections is witnessed by an *iterated* median cylinder  $\Xi = \chi_l(\chi_k(a, b), \text{cyl } c)$  whose four faces are exactly the three interpolations and one degeneracy, on the nose (Theorem 4.2); and every compatible sphere of corrections fills, by the sphere-filling theorem applied in  $\text{DM}_{n+k}$  — the correction system is *strictly contractible*. In the language of algebraic weak factorization systems [2], the strict layer supports pointed-endofunctor algebra structures in abundance (any section will do) but no monad-algebra structure; the contractible median system is the canonical pseudo-structure interpolating between the two.

The consequence for cubical type theory is worth stating plainly (Corollary 3.3): *any filling operation natural in the cube category must take values outside the strict layer*. Kan composition is not a convenience bolted onto the reparametrization calculus; it is forced by naturality itself. This gives the strict–weak divide studied throughout the series [4, 5, 6] a purely categorical root.

## 2. PRELIMINARIES

We freely use the setting of [7, 8]:  $\text{DM}_n$  is the lattice of monotone functions (up-sets) on the literal cube  $Q_n = \{0, 1\}^{2n}$ , coordinates in pairs  $(x_i, \neg x_i)$ ; *face points* have some pair in  $\{(0, 1), (1, 0)\}$ , *diagonal points* have all pairs in  $\{(0, 0), (1, 1)\}$ ; the boundary  $\partial g$  is the tuple of the  $2n$  substitutions  $x_i := 0, 1$ ; fibers of  $\partial$  are Boolean intervals  $[m, m \cup D] \cong 2^D$  with  $D$  an antichain of diagonal points;  $A_g$  denotes the antichain of minimal diagonal points of  $g$ , and  $r(g) = g \setminus A_g$  is the least element of the fiber of  $g$  (the defect formula). A substitution  $f : [m] \rightarrow [n]$  of  $\square_{\text{DM}}$  induces  $f^* : \text{DM}_n \rightarrow \text{DM}_m$  and the literal-cube map  $F : Q_m \rightarrow Q_n$ . A *section* is a function  $s$  from realizable boundaries to elements with  $\partial(s(B)) = B$ ; it is *natural* if  $s(\partial(f^*s(B))) = f^*s(B)$  for all  $f, B$  — i.e. the transport of a chosen filler is again chosen.

## 3. THE NO-GO THEOREM

**Theorem 3.1.** *No natural section exists, already for  $n \leq 2$  and already with respect to the endomorphisms of [2].*

*Proof.* Let  $m = \neg x \wedge y \wedge \neg y \in \text{DM}_2$ , the up-set of the single point  $\alpha = ((0, 1), (1, 1))$ . Its boundary fiber is a singleton: a diagonal point  $\delta$  could be added only if every face point above  $\delta$  already lay in  $m$ , and for each of the four diagonal points one checks a face point above it missing from  $m$  (e.g. for  $\delta = ((0, 0), (1, 1))$  the face point  $((1, 0), (1, 1))$  is not in  $m$ ). So any natural section satisfies  $s(\partial m) = m$ , and hence is forced on every transport:  $s(\partial(f^*m)) = f^*m$  for all  $f$ . Now

$$(x := y)^* m = y \wedge \neg y, \quad (x := x \wedge \neg x)^* m = (x \vee \neg x) \wedge y \wedge \neg y.$$

These two elements have the same boundary — their common fiber is the two-element interval with  $D = \{((0, 0), (1, 1))\}$ , of which  $y \wedge \neg y$  is the top and  $(x \vee \neg x) \wedge y \wedge \neg y$  the bottom — and they are distinct. Two different values are forced at one boundary; contradiction.  $\square$

**Remark 3.2.** The proof is the coherence dichotomy of [8] weaponized: the diagonal substitution reaches the invisible diagonal point (its  $F$  maps a face point onto it), the connective collapse does not, and a *unique* filler upstream leaves no freedom to absorb the difference downstream. Machine census (§5): unique fillers force values on 42 of the 54 non-singleton fibers of  $\text{DM}_2$ , with 142 pairwise conflicts of forced values; and independently of the forcing argument, an exhaustive backtracking search over all sections, constrained by 121 endo-substitutions of [2], finds no solution. The no-go is thus robust: it does not hinge on one clever element, and it needs neither higher cubes nor degeneracies.

**Corollary 3.3.** *Any filling operation for De Morgan cubical sets that is natural in  $\square_{\text{DM}}$  — in particular, any algebraic fibrancy structure on the representables, and any functorial fibrant replacement — takes some values outside the strict layer. Equivalently: in any cubical type theory over the De*

*Morgan interval whose filling operation is substitution-stable, fillers cannot all be reparametrizations; a weak layer is categorically necessary.*

*Proof.* A strict-valued natural filling operation restricts, on representables, to a natural section, which Theorem 3.1 forbids.  $\square$

#### 4. THE MEDIAN CORRECTION STRUCTURE

If no filler can be chosen, all fillers must be carried — with the identifications between them. The companion papers supply the identifications; here we show they are computed by a graded Boolean median and compose *on the nose* via iterated medians.

**Proposition 4.1** (fiber coordinates). *Let  $[m, m \cup D]$  be a fiber and  $\varphi = m \cup S$ ,  $\varphi' = m \cup S'$  with  $S, S' \subseteq D$ . Then the median interpolation in one fresh generator  $k$  is, as an up-set of  $Q_{n+1}$ ,*

$$\begin{aligned} \chi(\varphi, \varphi') &= \text{cyl}(m \cup (S \cap S')) \\ &\cup \{(\alpha, \kappa) : \alpha \in S \setminus S', \kappa \in \neg k\} \cup \{(\alpha, \kappa) : \alpha \in S' \setminus S, \kappa \in k\}, \end{aligned}$$

*i.e. the cylinder on the common part and the symmetric difference graded by the two half-cylinders of  $k$ : the Boolean median of  $S$ ,  $S'$  and the  $k$ -grading, computed pointwise on  $D$ .*

*Proof.* Evaluate  $(\varphi \wedge \neg k) \vee (k \wedge \varphi') \vee (\varphi \wedge \varphi')$  at a point  $(\alpha, \kappa)$ : for  $\alpha \in m \cup (S \cap S')$  the third disjunct fires for all  $\kappa$ ; for  $\alpha \in S \setminus S'$  only the first, i.e.  $\kappa \in \neg k$ ; for  $\alpha \in S' \setminus S$  only the second. (Machine-checked on all pairs of a rank-two fiber of  $\text{DM}_2$ .)  $\square$

**Theorem 4.2** (coherence by iterated medians). *Let  $a, b, c$  lie in one fiber. Then the iterated median cylinder*

$$\Xi := \chi_l(\chi_k(a, b), \text{cyl}_k(c)) \in \text{DM}_{n+2}$$

*has faces, on the nose:*

$$\Xi|_{l=0} = \chi_k(a, b), \quad \Xi|_{l=1} = \text{cyl}_k(c), \quad \Xi|_{k=0} = \chi_l(a, c), \quad \Xi|_{k=1} = \chi_l(b, c).$$

*More generally, every compatible sphere assembled from interpolations and cylinders over one fiber fills in  $\text{DM}_{n+j}$ : the correction system is strictly contractible.*

*Proof.* The four face identities are absorption computations from Proposition 4.1 (each face substitutes an endpoint into one grading; machine-checked exactly on a rank-two fiber). For the general statement, observe that a compatible sphere of corrections over a fiber has, at each original face of  $Q_n$ , the constant value of the common boundary — so it is a compatible sphere in  $\text{DM}_{n+j}$ , and the sphere-filling theorem of [7, Thm. 4.1] provides a filler, whose defect freedom is again a Boolean fiber. Contractibility at every level follows by induction, exactly as in the strict-contractibility analysis of [5].  $\square$

**Remark 4.3** (the pseudo-structure). In the terminology of algebraic weak factorization systems [2], generated in  $\widehat{\square_{\text{DM}}}$  by the boundary inclusions  $\partial\square^n \hookrightarrow \square^n$ : any section equips the representables with an algebra structure for the *pointed endofunctor* of the associated one-step construction, but Theorem 3.1 forbids the compatibilities required of a *monad* algebra valued in strict fillers. The data that does exist naturally is: the full Boolean fiber  $2^D$  of fillers (structure, not property), the defect antichain measuring every transport [8], and the strictly contractible median system of Theorem 4.2 identifying all choices coherently. This is precisely a fibered, strictly contractible *pseudo*-algebra — and the no-go says pseudo is optimal. Steiner’s anticipation that thin fillers in weak settings “exist but not necessarily uniquely” [3] is thus realized in the sharpest form: existence, canonical non-uniqueness, and provable impossibility of natural selection.

## 5. MACHINE VERIFICATION

Three computations back the theorems (scripts accompany the artifact). (1) *Forcing census*: over  $\text{DM}_2$ , the unique fillers (singleton fibers) force values, via endo-substitutions from a generating set of 121, on 42 of the 54 non-singleton fibers, with 142 pairwise conflicts; the specific core of Theorem 3.1 is one of them, chosen for the shortness of its by-hand verification. (2) *Exhaustive search*: a backtracking search with constraint propagation over all sections —  $2^{38} \cdot 4^{16}$  assignments for the 38 two-element and 16 four-element fibers — confirms unsatisfiability against the same 121 substitutions, independently of the forcing argument. (3) *Median identities*: the fiber-coordinate formula of Proposition 4.1 and all four face identities of Theorem 4.2 were verified exactly on a rank-two fiber of  $\text{DM}_2$ . We remark that (1) alone proves the theorem, (2) alone re-proves it, and the printed proof needs neither.

## 6. OUTLOOK

Two directions. First, the *quantitative geography of the obstruction*: the forcing graph — boundaries as vertices, forced transports as edges — was here used only for unsatisfiability, but its structure (which fibers are forced from how many unique fillers, and how conflicts distribute over the defect rank) should refine the coherence dichotomy of [8] into a cohomological invariant of the strict layer; the Kleene case [9], with its two polar fibers, is the natural test bed, and we expect its forcing graph to be consistent away from the poles. Second, the *converse direction*: Theorem 3.1 shows natural filling requires values beyond reparametrizations; how far beyond? The minimal natural filling operations valued in Kan composites — and whether the CCHM hcomp is initial among them in a precise sense — is, we believe, the right formulation of the “value of the weak layer” question raised across this series, and the strictly contractible median system of §4 is the comparison datum such a formulation needs.

Artifact. Scripts and the machine-checked companions are available in the artifact repository:

<https://github.com/r-shibusawa/cubical-strict-j>, archived at DOI 10.5281/zenodo.21405961.

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